Source: Sahil Bhola, Self-contained Proof for a Mixed-precision Block Jacobi Algorithm.

Selected deliverable: the full active result labeled theorem:cholesky_solve in main.tex (lines 550-610).

The original mathematical proof is available in paper.tex. The selected statement below is authoritative.
Earlier statements and notation supply context. A referenced result may be proved as a helper. Do not assume a
desired algorithmic bound or an unproved paper lemma as an added hypothesis.

SELECTED STATEMENT (verbatim LaTeX, proof omitted)
```
\begin{theorem}[Cholesky Solve]\label{theorem:cholesky_solve}
Let $\mat{a}\ireal{n\times n}$ be symmetric positive definite and let
$\vect{b}\ireal{n}$. Suppose that $\mat{a}\vect{x}=\vect{b}$ is solved in
floating-point arithmetic by computing an upper-triangular Cholesky factor
$\widehat{\mat{r}}$, followed by forward and backward substitution. Assume
that the factorization completes successfully, no overflow or underflow
occurs, and $(n+1)\urd<1$. Then the computed solution
$\hat{\vect{x}}$ satisfies
\begin{align}
    (\mat{a}+\Delta\mat{a})\hat{\vect{x}}=\vect{b},
\end{align}
where
\begin{align}
    \abs{\Delta\mat{a}}
    &\leq
    \left(\gamma_{n+1}+2\gamma_n+\gamma_n^2\right)
    \abs{\widehat{\mat{r}}^T}\abs{\widehat{\mat{r}}}
    \\
    &\leq
    \left(3\gamma_{n+1}+\gamma_n^2\right)
    \abs{\widehat{\mat{r}}^T}\abs{\widehat{\mat{r}}}.
\end{align}
The inequalities are interpreted componentwise.
\end{theorem}
```

SECTION CONTEXT
```
\subsection{Cholesky Factorization}\label{sec:cholesky_factorization}
To solve a linear system of equations via Cholesky factorization, we need to compute the Cholesky factors given
by the recurrance relation
\begin{align}
    \mat{r}_{i,j} = \left(\mat{a}_{i,j} - \sum_{k=1}^{i-1} \mat{r}_{k,i}\mat{r}_{k,j}\right) / \mat{r}_{i,i},
\qquad i = 1,\cdots, j - 1 \\
   \mat{r}_{j,j} =  \left(\mat{a}_{j,j} - \sum_{k=1}^{j-1} \mat{r}_{k,j}^2 \right)^{1/2}, \qquad j = 1,\dots,n.
\end{align}
In the follwing result, we present the backward error analysis for performing Cholesky factorization, re-
deriving~\cite[Theorem 10.4]{higham2002accuracy}.
```

[Earlier statement and proof omitted here.]

```
Now, using~\cref{theorem:solution_of_a_traingular_system,theorem:cholesky_factorization} we can obtain the
backward error bounds for solving a linear system of equations via Cholesky factorization.
```

EARLIER STATEMENTS FOR CONTEXT
```
\begin{theorem}[Solution of a Triangular System]\label{theorem:solution_of_a_traingular_system}
Let $\mat{t}\ireal{n\times n}$ be a nonsingular triangular matrix and let
$\vect{b}\ireal{n}$. Suppose that $\mat{t}\vect{x}=\vect{b}$ is solved by
forward substitution when $\mat{t}$ is lower triangular, or by backward
substitution when $\mat{t}$ is upper triangular. Assume that no overflow
or underflow occurs and that $n\urd<1$. Then the computed solution
$\hat{\vect{x}}$ satisfies
\begin{align}
    (\mat{t}+\Delta\mat{t})\hat{\vect{x}}=\vect{b},
    \qquad
    \abs{\Delta\mat{t}}\leq\gamma_n\abs{\mat{t}},
\end{align}
where the inequality is interpreted componentwise,
$\Delta\mat{t}$ has the same triangular structure as $\mat{t}$, and
$\gamma_n=n\urd/(1-n\urd)$.
\end{theorem}

\begin{theorem}[Cholesky Factorization]\label{theorem:cholesky_factorization}
Let $\mat{a}\ireal{n\times n}$ be symmetric positive definite, and suppose
that the Cholesky algorithm applied to $\mat{a}$ completes successfully in
floating-point arithmetic, producing an upper-triangular factor
$\widehat{\mat{r}}$. Assume that no overflow or underflow occurs and that
$(n+1)\urd<1$. Then there exists a symmetric perturbation $\Delta\mat{a}$
such that
\begin{align}
    \widehat{\mat{r}}^{T}\widehat{\mat{r}}
    &=\mat{a}+\Delta\mat{a},
    &
    \abs{\Delta\mat{a}}
    &\leq
    \gamma_{n+1}
    \abs{\widehat{\mat{r}}^{T}}
```

```
    \abs{\widehat{\mat{r}}},
\end{align}
where the inequality is interpreted componentwise and
$\gamma_{n+1}=(n+1)\urd/(1-(n+1)\urd)$.
\end{theorem}

COMMON ROUNDING MODEL AND ERROR CONSTANTS
\section{Floating-Point Model}
We assume the standard floating-point model~\cite{higham2002accuracy}: for every elementary
operation $\circ \in \{+,-,\times,/\}$,
\begin{equation}
    \fl(a \circ b) = (a \circ b)(1+\delta),
    \qquad
    \abs{\delta} \le \urd,
    \label{eq:fp-model}
\end{equation}
where $\urd$ is the unit roundoff and $\delta$ is the relative roundoff error.
Using~\cref{eq:fp-model} for numerical analysis we typicall obtain the product
\begin{align}
    \prod_{i=1}^n (1 + \delta_i)^{\rho_i} \define 1 + \theta_n, \qquad \rho = \pm 1,
    \label{eq:theta_n}
\end{align}
where $n$ denotes the number of accumulated roundoff error terms.
For numerical analysis, we seek reliable and sharp bounds for $\theta_n$.


\section{Deterministic Rounding Error Analysis}
The deterministic rounding error analysis proposes the following problem-size dependent constant
\begin{align}
\gamma_{n} \define \frac{n\urd}{1 - n\urd},
\label{eq:drea}
\end{align}
for $n\urd < 1$.
Here, $n$ is the number of accumulated error terms~\cite{higham2002accuracy}.
The problem-size dependent constant is monotonic, that is, $\gamma_{n+1} > \gamma_{n}$ for $\urd > 0$.
Performing backward/forward error analysis using~\cref{eq:drea} we obtain
\begin{align}
    \abs{\theta_n} \leq \gamma_n.
\end{align}

SOURCE NOTATION MACROS
\newcommand{\urd}{\ensuremath{\mathrm{u}}}
\newcommand{\define}{\ensuremath{\triangleq}}
\newcommand{\vect}[1]{\mathbf{\MakeLowercase{#1}}}
\newcommand{\mat}[1]{\mathbf{\MakeUppercase{#1}}}
\newcommand{\ireal}[1]{\in\mathbb{R}^{#1}}
\newcommand{\norm}[2][]{\left\|#2\right\|_{#1}}
\DeclareMathOperator{\fl}{fl}
\newcommand{\R}{\mathbb{R}}
\newcommand{\abs}[1]{\left|#1\right|}
\newcommand{\ceil}[1]{\left\lceil #1 \right\rceil}
\newcommand{\widehatgamma}[1]{\widehat{\gamma}_{#1}}
```